

第三章

§3 常系数线性微分方程组的解法.

一阶常系数线性方程组的基本解法.

一、转为二阶常系数线性方程.

eg1.
$$\begin{cases} \frac{dx}{dt} = 5x - 2y + e^{3t} \\ \frac{dy}{dt} = 6x - 2y. \end{cases} \xrightarrow{\text{消去 } y} \begin{cases} 2y = 5x - \frac{dx}{dt} + e^{3t} \\ \frac{d}{dt}(2y) = 12x - 2(2y). \end{cases}$$

代入, 有
$$\frac{d}{dt}(5x - \frac{dx}{dt} + e^{3t}) = 12x - 2(5x - \frac{dx}{dt} + e^{3t}).$$

$$5 \frac{dx}{dt} - \frac{d^2x}{dt^2} + 3e^{3t} = 12x - 10x + 2 \frac{dx}{dt} - 2e^{3t}.$$

$$\frac{d^2x}{dt^2} - 3 \frac{dx}{dt} + 2x = 5e^{3t}.$$

$\alpha=3$ 不是 $r^2 - 3r + 2 = 0$ 的根.

设 $\tilde{x} = ke^{3t}$. 代入有 $9k - 9k + 2k = 5$, $k = \frac{5}{2}$.

又 $\frac{d^2x}{dt^2} - 3 \frac{dx}{dt} + 2x = 0$ 通解为 $x = C_1 e^t + C_2 e^{2t}$.

则原方程通解为 $x = C_1 e^t + C_2 e^{2t} + \frac{5}{2} e^{3t}$.

则 $y = 2C_1 e^t + \frac{3}{2} C_2 e^{2t} + 3e^{3t}$.

二、一阶常系数线性齐次方程组的特征向量方法.

回顾: 矩阵特征值和特征向量.

λ 是 $|\lambda E - A| = 0$ 的根. α 是 $(\lambda E - A)\alpha = 0$ 的非零解.

eg. $A = \begin{bmatrix} 5 & -2 \\ 6 & -2 \end{bmatrix}$. $\lambda E - A = \begin{bmatrix} \lambda - 5 & 2 \\ -6 & \lambda + 2 \end{bmatrix}$.

k 重特征根 λ_0 .

有解:

$$(\vec{v}_0 + \frac{t}{1!} \vec{v}_1 + \dots + \frac{t^{k-1}}{(k-1)!} \vec{v}_{k-1}) e^{\lambda_0 t}$$

其中 $(A - \lambda_0 E)^k \vec{v}_0 = 0$

$$(A - \lambda_0 E) \vec{v}_i = \vec{v}_{i+1}.$$

$$|\lambda E - A| = (\lambda - 5)(\lambda + 2) + 12 = \lambda^2 - 3\lambda + 2 = 0.$$

$$\lambda_1 = 1, \lambda_2 = 2.$$

$$\left[\begin{array}{c} \left(\begin{array}{cc} -4 & 2 \\ -6 & 3 \end{array} \right) \alpha_1 = 0, \alpha_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \left(\begin{array}{cc} -3 & 2 \\ -6 & 4 \end{array} \right) \alpha_2 = 0, \alpha_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \end{array} \right]$$

· 特征向量法求解微分方程的基本思想.

$$\begin{cases} \frac{dx}{dt} = a_{11}x + a_{12}y \\ \frac{dy}{dt} = a_{21}x + a_{22}y \end{cases} \xrightarrow{\text{猜想}} \begin{cases} x = k_1 e^{\lambda t} \\ y = k_2 e^{\lambda t} \end{cases}$$

$$\Rightarrow \begin{cases} k_1 \lambda e^{\lambda t} = a_{11} k_1 e^{\lambda t} + a_{12} k_2 e^{\lambda t} \\ k_2 \lambda e^{\lambda t} = a_{21} k_1 e^{\lambda t} + a_{22} k_2 e^{\lambda t} \end{cases} \Rightarrow \begin{cases} k_1 \lambda = a_{11} k_1 + a_{12} k_2 \\ k_2 \lambda = a_{21} k_1 + a_{22} k_2 \end{cases}$$

$$\Rightarrow \lambda \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} \Rightarrow \begin{pmatrix} \lambda - a_{11} & -a_{12} \\ -a_{21} & \lambda - a_{22} \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = 0.$$

$\Rightarrow \lambda$ 是 $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$ 特征值, $\begin{pmatrix} k_1 \\ k_2 \end{pmatrix}$ 为相应特征向量.

$$\Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} k_1 e^{\lambda t} \\ k_2 e^{\lambda t} \end{pmatrix} = \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} e^{\lambda t} \text{ 为解.}$$

eg2. $\begin{cases} \frac{dx}{dt} = 5x - 2y \\ \frac{dy}{dt} = 6x - 2y \end{cases}$

(1) 写出矩阵, 求出特征值, 特征向量.

$$\begin{vmatrix} \lambda - 5 & 2 \\ -6 & \lambda + 2 \end{vmatrix} = \lambda^2 - 3\lambda + 2 = 0. \lambda_1 = 1, \alpha_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad \lambda_2 = 2, \alpha_2 = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$$

12) 写出特解 $\begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t, \begin{pmatrix} 2 \\ 3 \end{pmatrix} e^{2t}$

13) 写出通解.

$$\begin{pmatrix} x \\ y \end{pmatrix} = C_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} e^t + C_2 \begin{pmatrix} 2 \\ 3 \end{pmatrix} e^{2t} = \begin{pmatrix} C_1 e^t + 2C_2 e^{2t} \\ 2C_1 e^t + 3C_2 e^{2t} \end{pmatrix}$$

方法的归纳和提炼.

$$\begin{cases} \frac{dx}{dt} = a_{11}x + a_{12}y \\ \frac{dy}{dt} = a_{21}x + a_{22}y \end{cases} \Rightarrow \frac{d}{dt} \vec{x} = A \vec{x} \quad (\text{其中 } A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \vec{x} = \begin{pmatrix} x \\ y \end{pmatrix}).$$

$\Rightarrow$ 求出 A 所有特征值 λ_1, λ_2 及对应的线性无关特征向量 $\vec{v}_1, \vec{v}_2$.

$\Rightarrow$ 特解为 $\vec{x}_1 = \vec{v}_1 e^{\lambda_1 t}, \vec{x}_2 = \vec{v}_2 e^{\lambda_2 t}$.

$\Rightarrow$ 通解为 $\vec{x} = C_1 \vec{x}_1 + C_2 \vec{x}_2 = C_1 \vec{v}_1 e^{\lambda_1 t} + C_2 \vec{v}_2 e^{\lambda_2 t}$

以上方法可推广至三元以上.

eg.
$$\begin{cases} \frac{dx}{dt} = 3x - y + z \\ \frac{dy}{dt} = -x + 5y - z \\ \frac{dz}{dt} = x - y + 3z \end{cases}$$

$$A = \begin{pmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{pmatrix}, \quad |\lambda E - A| = \begin{vmatrix} \lambda - 3 & 1 & -1 \\ 1 & \lambda - 5 & 1 \\ -1 & 1 & \lambda - 3 \end{vmatrix}$$

$$\lambda_1 = 2, \lambda_2 = 3, \lambda_3 = 6.$$

$$\vec{v}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \vec{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \vec{v}_3 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$$

特解为 $\vec{v}_1 e^{\lambda_1 t}, \vec{v}_2 e^{\lambda_2 t}, \vec{v}_3 e^{\lambda_3 t}$

通解为 $\vec{x} = C_1 \cdot \vec{v}_1 e^{\lambda_1 t} + C_2 \cdot \vec{v}_2 e^{\lambda_2 t} + C_3 \cdot \vec{v}_3 e^{\lambda_3 t}$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = C_1 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} e^{2t} + C_2 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} e^{3t} + C_3 \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix} e^{6t}.$$

回忆 实矩阵的复特征值与特征向量.

若实阵 A 有复特征值 λ , 对应特征向量为 $\vec{z}$.

$$\Rightarrow A\vec{z} = \lambda\vec{z}, \Rightarrow \overline{A\vec{z}} = \overline{\lambda\vec{z}} \quad (\text{取共轭}).$$

$$\Rightarrow A\overline{\vec{z}} = \overline{\lambda}\overline{\vec{z}}. \Rightarrow \overline{\vec{z}} \text{ 是 } A \text{ 的属于 } \overline{\lambda} \text{ 的特征向量}$$

结论: 实阵 A 的复特征值是以共轭形式出现的, 特征向量也可取成共轭.

eg. $\begin{cases} \frac{dx}{dt} = -y \\ \frac{dy}{dt} = x \end{cases}$

$$A = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, | \lambda E - A | = \begin{vmatrix} \lambda & 1 \\ -1 & \lambda \end{vmatrix} = \lambda^2 + 1$$

$$\lambda_1 = i, \lambda_2 = -i. \quad \vec{v}_1 = \begin{pmatrix} i \\ 1 \end{pmatrix}, \vec{v}_2 = \begin{pmatrix} -i \\ 1 \end{pmatrix}$$

$$\vec{x}_1 = \begin{pmatrix} i \\ 1 \end{pmatrix} e^{it} = \left[\begin{pmatrix} 0 \\ 1 \end{pmatrix} + i \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right] \cdot (\cos t + i \sin t)$$

$$= \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} + i \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$$

$$\vec{x}_2 = \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} - i \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} \quad \swarrow \text{共轭}$$

特解: $\frac{1}{2}(\vec{x}_1 + \vec{x}_2) = \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix}$ $\frac{1}{2i}(\vec{x}_1 - \vec{x}_2) = \begin{pmatrix} \cos t \\ \sin t \end{pmatrix}$

通解: $\begin{pmatrix} x \\ y \end{pmatrix} = C_1 \begin{pmatrix} -\sin t \\ \cos t \end{pmatrix} + C_2 \begin{pmatrix} \cos t \\ \sin t \end{pmatrix} = \begin{pmatrix} -C_1 \sin t + C_2 \cos t \\ C_1 \cos t + C_2 \sin t \end{pmatrix}$

三、一阶常系数线性非齐次方程组的常数变异法.

预备知识——记号与运算

(1) 约定 $[a_{ij}(t)]' = [a'_{ij}(t)]$ (矩阵的导数)

(2) 记 $\begin{pmatrix} x \\ y \end{pmatrix} = \vec{x}$ (解向量), 且 $\vec{x}' = \frac{d}{dt} \vec{x} = \begin{pmatrix} x' \\ y' \end{pmatrix}$ (解向量的导数)

(3) 若 $\vec{x}_1, \vec{x}_2$ 为 $\frac{d}{dt} \vec{x} = A\vec{x}$ 的解 $\Rightarrow \frac{d}{dt} \vec{x}_1 = A\vec{x}_1, \frac{d}{dt} \vec{x}_2 = A\vec{x}_2$.

(4) 记 $(\vec{x}_1 \ \vec{x}_2) = X$ (称为基本解矩阵) $\begin{bmatrix} C_1(t) \\ C_2(t) \end{bmatrix} = \vec{C}(t)$.

(5) $\frac{d}{dt} X = \frac{d}{dt} (\vec{x}_1 \ \vec{x}_2) = \begin{pmatrix} \frac{d}{dt} \vec{x}_1 & \frac{d}{dt} \vec{x}_2 \end{pmatrix} = (A\vec{x}_1 \ A\vec{x}_2) = A(\vec{x}_1 \ \vec{x}_2) = AX$.

(6) $\frac{d}{dt} [X\vec{C}(t)] = \frac{d}{dt} \left\{ (\vec{x}_1 \ \vec{x}_2) \begin{bmatrix} C_1(t) \\ C_2(t) \end{bmatrix} \right\} = \frac{d}{dt} [C_1(t)\vec{x}_1 + C_2(t)\vec{x}_2]$
 $= [C_1'(t)\vec{x}_1 + C_2'(t)\vec{x}_2] + [C_1(t)\vec{x}_1' + C_2(t)\vec{x}_2']$
 $= (\vec{x}_1 \ \vec{x}_2) \begin{bmatrix} C_1'(t) \\ C_2'(t) \end{bmatrix} + (\vec{x}_1' \ \vec{x}_2') \begin{bmatrix} C_1(t) \\ C_2(t) \end{bmatrix} = X\vec{C}'(t) + X'\vec{C}(t)$

归纳 $\frac{d}{dt} X = AX$; $\frac{d}{dt} [X\vec{C}(t)] = X\vec{C}'(t) + X'\vec{C}(t)$ (函数之积的导数性质)

1. 求出非齐次的特解.

$\begin{cases} \frac{dx}{dt} = a_{11}x + a_{12}y + f_1(t) \\ \frac{dy}{dt} = a_{21}x + a_{22}y + f_2(t) \end{cases}$ 写成 $\frac{d}{dt} \vec{x} = A\vec{x} + \vec{f}(t)$

其中 $A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$, $\vec{x} = \begin{pmatrix} x \\ y \end{pmatrix}$, $\vec{f}(t) = \begin{pmatrix} f_1(t) \\ f_2(t) \end{pmatrix}$

若齐次通解为 $\vec{x} = C_1 \vec{x}_1 + C_2 \vec{x}_2$.

设非齐次特解为 $\vec{x}^* = C_1(t) \vec{x}_1 + C_2(t) \vec{x}_2 = (\vec{x}_1 \ \vec{x}_2) \begin{pmatrix} C_1(t) \\ C_2(t) \end{pmatrix} = X \vec{C}(t)$

$$\Rightarrow \frac{d}{dt} (X \vec{C}(t)) = A X \vec{C}(t) + \vec{f}(t)$$

$$\Rightarrow X \cdot \frac{d}{dt} \vec{C}(t) + \frac{d}{dt} X \cdot \vec{C}(t) = A X \vec{C}(t) + \vec{f}(t)$$

$$\Rightarrow X \cdot \frac{d}{dt} \vec{C}(t) = \vec{f}(t) \Rightarrow X \vec{C}'(t) = \vec{f}(t) \Rightarrow \vec{C}'(t) = X^{-1} \vec{f}(t).$$

2. 写出通解.

结论: $\frac{d}{dt} \vec{x} = A \vec{x}$ 通解: $\vec{x} = C_1 \vec{x}_1 + C_2 \vec{x}_2$

$\frac{d}{dt} \vec{x} = A \vec{x} + \vec{f}(t)$ 特解: $\vec{x}^* = C_1(t) \vec{x}_1 + C_2(t) \vec{x}_2$.

$\frac{d}{dt} \vec{x} = A \vec{x} + \vec{f}(t)$ 通解: $\vec{x} = C_1 \vec{x}_1 + C_2 \vec{x}_2 + \vec{x}^*$

其中 $X = (\vec{x}_1 \ \vec{x}_2)$ $\vec{C}(t) = \begin{pmatrix} C_1(t) \\ C_2(t) \end{pmatrix}$ $X \vec{C}'(t) = \vec{f}(t)$

以上方法可推广到三元以上.

eg5. $\begin{cases} \frac{dx}{dt} = 5x - 2y + e^{3t} \\ \frac{dy}{dt} = 6x - 2y \end{cases}$, 其中 $\begin{cases} \frac{dx}{dt} = 5x - 2y \\ \frac{dy}{dt} = 6x - 2y \end{cases}$ 特解 $\vec{x}_1 = \begin{pmatrix} e^t \\ 2e^t \end{pmatrix}$, $\vec{x}_2 = \begin{pmatrix} 2e^{2t} \\ 3e^{2t} \end{pmatrix}$. 通解 $\begin{pmatrix} x \\ y \end{pmatrix} = C_1 \vec{x}_1 + C_2 \vec{x}_2$.

改写成矩阵形式: $\frac{d}{dt} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5 & -2 \\ 6 & -2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} e^{3t} \\ 0 \end{pmatrix}$

$A = \begin{pmatrix} 5 & -2 \\ 6 & -2 \end{pmatrix}$, $\vec{x} = \begin{pmatrix} x \\ y \end{pmatrix}$, $\vec{f}(t) = \begin{pmatrix} e^{3t} \\ 0 \end{pmatrix}$.

对应齐次的通解 $\begin{pmatrix} x \\ y \end{pmatrix} = C_1 \vec{x}_1 + C_2 \vec{x}_2$, $X = (\vec{x}_1 \ \vec{x}_2) = \begin{pmatrix} e^t & 2e^{2t} \\ 2e^t & 3e^{2t} \end{pmatrix}$

非齐次的特解 $\vec{x}^* = C_1(t) \vec{x}_1 + C_2(t) \vec{x}_2$

$$X \begin{bmatrix} C_1'(t) \\ C_2'(t) \end{bmatrix} = \vec{f}(t) \Rightarrow \begin{bmatrix} e^t & 2e^{2t} \\ 2e^t & 3e^{2t} \end{bmatrix} \begin{bmatrix} C_1' \\ C_2' \end{bmatrix} = \begin{bmatrix} e^{3t} \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} e^t C_1' \\ e^{2t} C_2' \end{bmatrix} = \begin{bmatrix} e^{3t} \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} e^t C_1' \\ e^{2t} C_2' \end{bmatrix} = \begin{bmatrix} -3e^{3t} \\ 2e^{3t} \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} C_1' \\ C_2' \end{bmatrix} = \begin{bmatrix} -3e^{2t} \\ 2e^t \end{bmatrix} \Rightarrow \begin{bmatrix} C_1(t) \\ C_2(t) \end{bmatrix} = \begin{bmatrix} -1.5e^{2t} \\ 2e^t \end{bmatrix}$$

非齐次特解 $\vec{x}^* = C_1(t) \vec{x}_1 + C_2(t) \vec{x}_2$

$$= -1.5e^{2t} \begin{pmatrix} e^t \\ e^{2t} \end{pmatrix} + 2e^t \begin{pmatrix} 2e^{2t} \\ 3e^{2t} \end{pmatrix}$$

$$= \begin{pmatrix} 2.5e^{3t} \\ 3e^{3t} \end{pmatrix}$$

齐次通解 $\begin{pmatrix} x \\ y \end{pmatrix} = C_1 \vec{x}_1 + C_2 \vec{x}_2 = \begin{pmatrix} C_1 e^t + 2C_2 e^{2t} \\ 2C_1 e^t + 3C_2 e^{2t} \end{pmatrix}$

非齐次通解 $\vec{x} = C_1 \vec{x}_1 + C_2 \vec{x}_2 + \vec{x}^* = \begin{pmatrix} C_1 e^t + 2C_2 e^{2t} + 2.5e^{3t} \\ 2C_1 e^t + 3C_2 e^{2t} + 3e^{3t} \end{pmatrix}$

另解: $\begin{cases} \frac{dx}{dt} = 5x - y + e^{3t} \\ \frac{dy}{dt} = 6x - 2y \end{cases} \Rightarrow \begin{cases} y = 5x - \frac{dx}{dt} + e^{3t} \\ \frac{d}{dt}(y) = 12x - 2(y) \end{cases}$

常系数线性方程组

消元法

特征向量法

常数变易法

求齐次/非齐次方程组通解.

求齐次方程组通解.

求非齐次方程组特解.

§1 微分方程组与线性微分方程组

一、基本概念

(1) 一阶微分方程组
$$\begin{cases} \frac{dx_1}{dt} = f_1(t, x_1, \dots, x_n) \\ \frac{dx_2}{dt} = f_2(t, x_1, \dots, x_n) \\ \vdots \\ \frac{dx_n}{dt} = f_n(t, x_1, \dots, x_n) \end{cases}$$

通解形式
$$\begin{cases} x_1 = \varphi_1(t, C_1, \dots, C_n) \\ x_2 = \varphi_2(t, C_1, \dots, C_n) \\ \vdots \\ x_n = \varphi_n(t, C_1, \dots, C_n) \end{cases}$$

初值条件 $x_1(t_0) = x_{10}, \dots, x_n(t_0) = x_{n0}.$

复习

一阶方程 $\frac{dy}{dx} = f(x, y)$

基本概念：线性 阶数 (两种) 齐次 特解/通解

基本类型：

$$\frac{dy}{dx} = \varphi(x) \psi(y) \quad \text{变量分离型}$$

$$\frac{dy}{dx} = g\left(\frac{y}{x}\right) \quad \text{变量型齐次}$$

$$\frac{dy}{dx} + p(x)y = f(x) \quad \text{一阶线性}$$

$$Mdx + Ndy = 0. \quad \text{全微分方程.}$$

须记忆：

$$\frac{dy}{dx} + p(x)y = f(x) \quad \text{通解:}$$

$$y = e^{-\int p(x)dx} \left[\int f(x) e^{\int p(x)dx} dx + C \right].$$

1. $(x^2 - 1)dy + (2xy - \cos x)dx = 0, \quad y(0) = 1.$

全微分.

2. $(3x^2 + 2xy - y^2)dx + (x^2 - 2xy)dy = 0.$

全微分.

3. $xy' + y = xe^x$

直接积分.

$$4. \quad xy' - y[\ln(xy) - 1] = 0, \quad y(1) = e^e.$$

观察、分组

$$xy' + y - y\ln(xy) = 0 \Rightarrow (xy)' - y\ln(xy) = 0.$$

$$\Rightarrow (xy)' = xy \ln(xy) \frac{1}{x} \Rightarrow \frac{d(xy)}{xy \ln(xy)} = \frac{1}{x} dx. \quad (\text{要讨论})$$

$$5. \quad \frac{dy}{dx} = \frac{1}{x - 2y}.$$

角色互换 $\frac{dx}{dy} = x - 2y. \quad \frac{dx}{dy} - x = -2y.$

$$6. \quad y' - (x+y)^2 = 0.$$

换元, $u' - 1 - u^2 = 0 \quad (u = x + y).$

$$7. \quad xy y' = x^2 + y^2, \quad y(e) = 2e.$$

法一: 变量型齐次, $y' = \frac{x}{y} + \frac{y}{x}.$

法二: 换元变成线性, $\frac{1}{2} x (y^2)' = x^2 + y^2$

$$\frac{1}{2} x u' = x^2 + u \quad (u = y^2).$$

二阶方程 $y'' = f(x, y, y')$.

可降阶型

$\left\{ \begin{array}{l} \text{二阶常系数线性} \\ \text{二阶变系数线性} \\ \text{变形及向高阶的推广.} \end{array} \right. \longrightarrow \begin{array}{l} \text{Euler 型} \\ \text{Liouville 法} \\ \text{常数变易法.} \end{array}$

二阶线性方程猜特解的方法

常见特解 $x^n, e^{\lambda x}$

eg. $x, x^2, \frac{1}{x}, \frac{1}{x+a}$.

曾经求解过的二阶方程, 收集猜到的特解.

1. $x^2 y'' - xy' + y = 0$.

$y_1 = x$.

2. $y'' + \frac{x}{1-x} y' - \frac{1}{1-x} y = 0$.

$y_1 = x$ 或 $y_1 = e^x$.